Code No: 117EE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, May/June - 2019
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Explain briefly about Associative Arrays. [2]
b) Write a short note on Disk Utility. [3]
c) What is a Regular File in UNIX System? [2]
d) Write a short note on "lseek". [3]
e) Differentiate between Threads and Processes. [2]
f) Write a short note on Environment Variables. [3]
g) Write a short note on IPC. [2]
h) Explain briefly about "popen" library function. [3]
i) What is a Socket? [2]
j) Write a short note on "fcntl" system call. [3]

PART-B

(50 Marks)

2. Explain in detail about Text Processing Utilities and Backup Utilities. [10]

OR

3.a) Write a detailed note on Shell Responsibilities.
b) Explain in detail about Shell Meta Characters. [5+5]

4. Explain in detail about following File Operations :
a) open b) creat c) read d) write [10]

OR

5. Explain in detail about File Ownership Commands. [10]

6. Explain in detail about Process Creation Mechanism. [10]

OR

7. Write a detailed note on following functions:
a) kill b) raise c) alarm d) pause [10]

8.a) Discuss about Kernel support for Message Queues.
b) Demonstrate client/server communication using message queues. [5+5]

OR

9. Explain in detail about IPC between unrelated processes using FIFO's. [10]

JNTUH USED 27-05-2019PM

10. Explain in detail about Socket System Calls for Connection Oriented and Connection less protocol. [10]

OR

11. Write a short note on API's for Shared Memory. [10]

--ooOoo--

JNTUH USED 27-05-2019PM